

Program: Diploma in Food Processing Technology	
Course Code: 4422	Course Title: Food Chemistry I
Semester: 4	Credits: 3
Course Category:	
Periods per week: 4 (L: 3T:1 P: 0)	Periods per semester: 60

Course Objectives:

1. To understand the quality changes in food components during processing and preservation
2. To implement the basic knowledge about food components acquired for producing various food products for different age groups
3. To impart knowledge in food toxicology and toxicants

Course Prerequisites:

Topic	Program/Course Name
Basic knowledge of chemistry	Applied chemistry

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Predict the the quality changes in food components during processing and preservation	15	Understanding
CO2	Explain the basic knowledge of food components for producing various food products for different age groups	15	Understanding
CO3	Illustrate the basic principles of food toxicology, classifications and the factor affecting toxicity.	15	Understanding
CO4	Describe the natural and derived food toxicants and their mode of action	15	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Predict the quality changes in food components during processing and preservation		
M1.01	Understand the Browning reactions, Dextrose Equivalent, Sugar alcohols	5	Understanding
M1.02	Explain Starch thickening & gelatinization, modified starches, resistant starch	5	Understanding
M1.03	Outline Rancidity and its types; Shortening power of fats, tenderization, emulsification	5	Understanding
Contents: Browning reactions- Caramelization, Maillard reaction, Dextrose Equivalent, Sugar alcohols; Starch- properties, thickening & gelatinization, modified starches, resistant starch, Rancidity and its types; Shortening power of fats, tenderization, emulsification			
CO2	Explain the basic knowledge of food components for producing various food products for different age groups		
M2.01	Discuss Basal metabolic rate- Calorific value of foods	4	Understanding
M2.02	Define Glycemic Index	3	Understanding
M2.03	Describe Biological value of protein	4	Understanding
M2.04	Review Protein Malnutrition- Obesity	3	Understanding
	Series Test –I	1	

Contents: Basal metabolic rate- Calorific value of food -balanced diet- Glycemic Index- Biological value of protein- Protein Malnutrition- Obesity			
CO3	Illustrate the basic principles of food toxicology, classifications and the factor affecting toxicity.		
M3.01	Define toxicology and its principles	5	Understanding
M3.02	Classify the food toxicants	5	Understanding
M3.03	List the factors affecting toxicity of compounds.	5	Understanding
Contents: Definition, scope, Principles of toxicology, Classification of food toxicants, factors affecting toxicity of compounds			
CO4	Describe the natural and derived food toxicants and their mode of action		
M4.01	Explain Natural Food toxicants	4	Understanding
M4.02	Discuss Derived Food toxicants	5	Understanding
M4.03	Outline Food carcinogen and mutagens and their mode of action	5	Understanding
	Series Test–II	1	
contents: Toxicants and allergens derived from plants, animals, marine algae, mushrooms etc. Microbial toxins, Toxicants generated during food processing and packaging such as nitrosamines, acrylamide, etc., food carcinogen and mutagens and their mode of action			

TEXT/REFERENCE

T/R	Book Title/Author
R1	Concon JM.1988. Food Toxicology - Principles & Concepts. Marcel Dekker.
R2	Steven T. 1989. Food Toxicology: A Perspective on Relative Risks
R3	Helferich, William and Carl K. Winter “Food Toxicology” CRC Press, 2001

R4	Shibamoto, Taka yuki and Leonard F.Bjeldanzes “Introduction to Food Toxicology” 2nd Edition.Academic Press, 2009.
R5	Shibamoto T. and Bjeldanes L., Introduction to Food Toxicology, Academic Press, Inc. San Diego, CA. ISBN 0-12640025-3 2. Deshpande, S.S. (2002).
R6	Handbook of Food Toxicology, Marcel Dekker Inc. NY ISBN 0- 8247-0760-5
R7	Stine, K. and Brown, T. 1996. Principles of Toxicology. CRC Press, Inc. Boca Raton, FL.
R8	Püssa, Tõnu. 2007. Principles of Food Toxicology. CRC Press, LLC. Boca Raton, FL. 5.

Online Resources

Sl.No	Website Link
1	https://foodscience.ucdavis.edu/academic-programs/undergraduate/courses/fst-128-food-toxicology
2	https://www.foodallergy.org/living-food-allergies/food-allergy-essentials/common-allergens
3	https://www.fda.gov/food/chemical-contaminants-pesticides/natural-toxins-food
4	https://www.who.int/news-room/fact-sheets/detail/natural-toxins-in-food
5	https://www.futurelearn.com/info/courses/nutrition-for-health/0/steps/213600